

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-5115
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

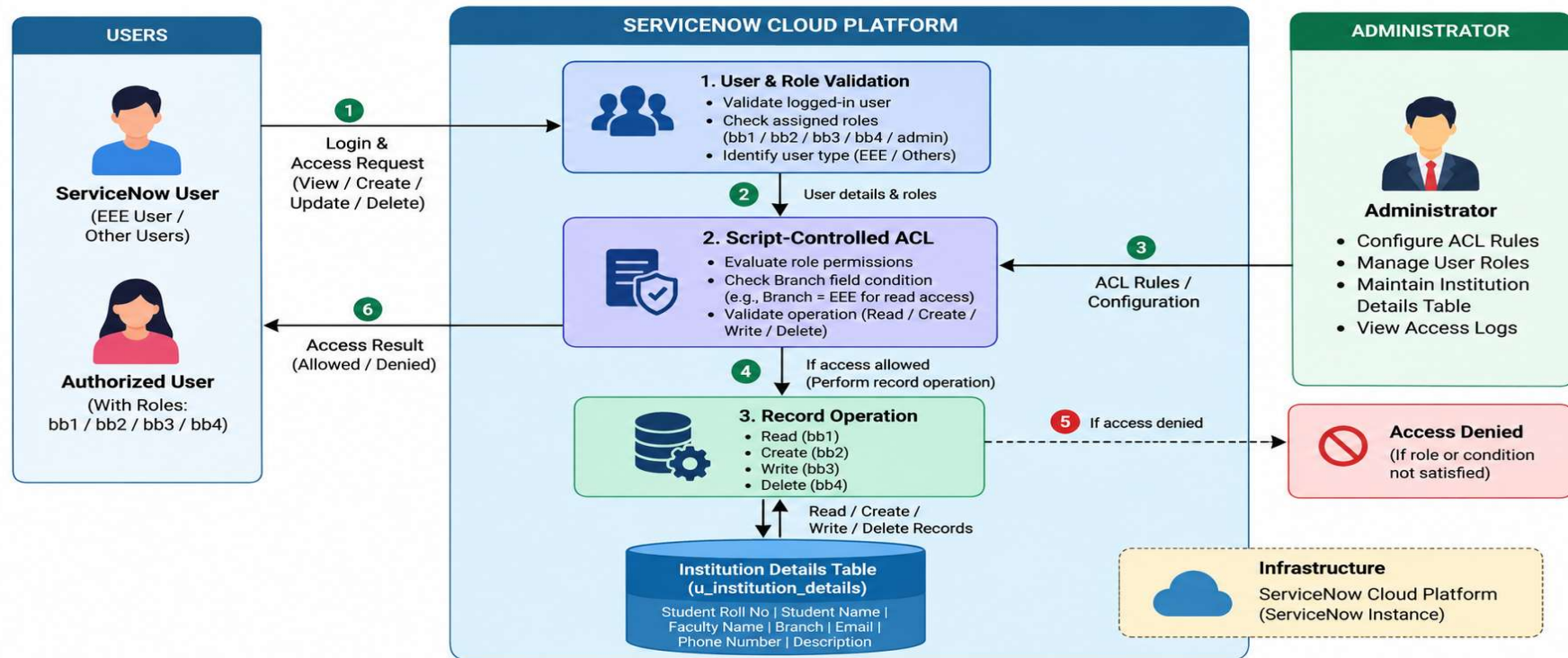


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides an interface for users and administrators to access and manage institution records.	ServiceNow UI
2.	Application Logic-1	Authenticates users and verifies their assigned roles.	ServiceNow User and Role Management
3.	Application Logic-2	Controls read access based on user roles and the Branch field condition.	Scripted ACL, JavaScript
4.	Application Logic-3	Controls Create, Write, and Delete operations based on assigned roles.	ServiceNow ACL Configuration
5.	Database	Stores institution and student information.	ServiceNow Platform Database
6.	Cloud Database	Manages application data within the ServiceNow platform environment.	ServiceNow-Managed Database
7.	File Storage	No separate file storage is required for this project.	Not Applicable
8.	External API-1	No external API integration is required.	Not Applicable
9.	External API-2	No additional external API integration is required.	Not Applicable
10.	Machine Learning Model	The project uses rule-based access control and does not require a machine learning model.	Not Applicable
11.	Infrastructure (Server / Cloud)	Hosts the application and executes ACL rules within the ServiceNow instance.	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source framework is required. The solution uses built-in ServiceNow features and server-side scripting.	ServiceNow Platform, JavaScript
2.	Security Implementations	Restricts record access using assigned roles, scripted ACLs, and the Branch field condition.	ServiceNow ACL, Roles (bb1, bb2, bb3, bb4)
3.	Scalable Architecture	Supports additional institution records and users within the ServiceNow instance.	ServiceNow Platform

S.No	Characteristics	Description	Technology
4.	Availability	Application availability depends on the availability of the ServiceNow instance.	ServiceNow Cloud Platform
5.	Performance	Evaluates user roles and configured ACL conditions before allowing record operations.	Server-Side ACL Scripts